MODEL BUILDING AND FORECASTING

Learning Outcome

- describe how ratio analysis and other techniques can be used to model and forecast earnings

Analysts often need to forecast future financial performance. For example, analysts' EPS forecasts and related equity valuations are widely followed by Wall Street. Analysts use data about the economy, industry, and company in arriving at a company's forecast. The results of an analyst's financial analysis, including common-size and ratio analyses, are integral to this process, along with the judgment of the analysts.

Based upon forecasts of growth and expected relationships among the financial statement data, the analyst can build a model to forecast future performance, which will be covered in later modules. In addition to budgets, pro forma financial statements are widely used in financial forecasting within companies, especially for use by senior executives and boards of directors. Last but not least, these budgets and forecasts are also used in presentations to credit analysts and others in obtaining external financing.

For example, based on a revenue forecast, an analyst may budget expenses based on expected common-size data. Forecasts of balance sheet and cash flow statements can be derived from expected ratio data, such as DSO. Forecasts are not limited to a single point estimate but should involve a range of possibilities. This can involve several techniques:

- **Sensitivity analysis**: Also known as "what if" analysis, sensitivity analysis shows the range of possible outcomes as specific assumptions are changed; this could, in turn, influence financing needs or investment in fixed assets.
- **Scenario analysis**: This type of analysis shows the changes in key financial quantities that result from given (economic) events, such as the loss of customers, the loss of a supply source, or a catastrophic event. If the list of events is mutually exclusive and exhaustive and the events can be assigned probabilities, the analyst can evaluate not only the range of outcomes but also standard statistical measures such as the mean and median value for various quantities of interest.
- **Simulation**: This is computer-generated sensitivity or scenario analysis based on probability models for the factors that drive outcomes. Each event or possible outcome is assigned a probability. Multiple scenarios are then run using the probability factors assigned to the possible values of a variable to determine an expected outcome for that variable.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{FA ROE} FINANCIAL ANALYSIS: DUPONT ANALYSIS

The Bloomberg FA ROE function allows you to view the full DuPont disaggregation of return on equity for any public company and see the effects of changes in its components.

> "FA <GO> provides comprehensive financial information for a company or equity index, including historical fundamental data and future estimates, so you can

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basis. The DuPont Analysis subtab, or FA ROE <GO>, allows you to view the decomposition of return of equity into its drivers." -Bloomberg

To explore this function, type FA ROE into the Bloomberg command line and press Enter or <GO>. Type a security—such as Sociedad Quimica y Minera de Chile SA (or SQM/B CI Equity)—into the top left amber field, and press Enter or <GO>.

In Millions of USD except Per Share	2016 Y	2017 Y	2018 Y	2019 Y	2020 Y	2021 Y	2022 Y
12 Months Ending	'31/2016	12/31/2017	12/31/2018	12/31/2019	12/31/2020	12/31/2021	12/31/2022
Tax Burden							
Net Inc to Comn/Pre-Tax Profit %	67.08	71.93	70.82	71.20	68.97	69.60	71.20
Adjustment Factor							
Normlzd Net Inc/Net Inc to Cmn	0.96	1.04	1.01	1.02	1.28	1.06	1.00
Interest Burden							
Pre-Tax Profit/EBIT %	87.84	92.49	91.77	84.53	73.93	91.11	98.38
Operating Margin							
EBIT/Revenue %	24.36	29.80	29.87	23.78	17.75	32.26	52.07
Asset Turnover							
Revenue/Avg Assets	0.44	0.51	0.53	0.43	0.38	0.48	1.20
Leverage Ratio							
Avg Assets/Avg Equity	1.93	1.92	2.00	2.15	2.26	2.24	2.21
Adjusted Return on Equity	11.61	20.13	20.71	13.66	10.04	23.38	96.87
5 Year Average Adj ROE	16.98	14.34	14.63	15.05	15.23	17.58	32.93
Payout Ratio	141.49	86.63	121.01	115.01	128.78	96.08	57.14
Sustainable Growth Rate	-5.04	2.58	-4.33	-2.00	-2.25	0.86	41.45

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Discussion

What do you want to discuss... (required)

MO

From practice Qs. Can somebody explain to me why they simply compare previous year to targeted figures and not taking into account what would happen if they did nothing? See below my answer vs their answer.

Brown Corporation had average days of sales outstanding of 19 days in the most recent fiscal year. Brown wants to improve its credit policies and collection practices and decrease its collection period in the next fiscal year to match the industry average of 15 days. Credit sales in the most recent fiscal year were $300 million, and Brown expects credit sales to increase to $390 million in the next fiscal year. To achieve Brown's goal of decreasing the collection period, the change in the average accounts receivable balance that must occur is closest to:

A. +USD0.41 million.

B. –USD0.41 million.

C. –USD1.22 million.

My answer:

Current AR = $300mil / 365days * 19 days = 15.62

If no change next year = $390mil / 365days * 19 days = 20.30

Target AR = $390mil / 365days * 15 days = 16.03

To achieve the 15 days target with a higher sales on credit of $390mil we would need to

20.30 – 16.03 = 4.27

Decrease the average AR balance by –4.27

Book's answer:

The average accounts receivable balances (actual and desired) must be calculated to determine the desired change. The average accounts receivable balance can be calculated as an average day's credit sales times the DSO. For the most recent fiscal year, the average accounts receivable balance is USD15.62 million [= (USD300,000,000/365) × 19]. The desired average accounts receivable balance for the next fiscal year is USD16.03 million (= (USD390,000,000/365) × 15). This is an increase of USD0.41 million (= 16.03 million – 15.62 million).

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